

Proof of the Clustered Hadwiger Conjecture

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Abstract—Hadwiger’s Conjecture asserts that every K_h -minor-free graph is properly $(h - 1)$ -colourable. We prove the following improper analogue of Hadwiger’s Conjecture: for fixed h , every K_h -minor-free graph is $(h - 1)$ -colourable with monochromatic components of bounded size. The number of colours is best possible regardless of the size of monochromatic components. It solves an open problem of Edwards, Kang, Kim, Oum and Seymour [SIAM J. Disc. Math. 2015], and concludes a line of research initiated in 2007. Similarly, for fixed $t \geq s$, we show that every $K_{s,t}$ -minor-free graph is $(s + 1)$ -colourable with monochromatic components of bounded size. The number of colours is best possible, solving an open problem of van den Heuvel and Wood [J. London Math. Soc. 2018]. We actually prove a single theorem from which both of the above results are immediate corollaries. For an excluded apex minor, we strengthen the result as follows: for fixed $t \geq s \geq 3$, and for any fixed apex graph X , every $K_{s,t}$ -subgraph-free X -minor-free graph is $(s + 1)$ -colourable with monochromatic components of bounded size. The number of colours is again best possible.

1. Introduction

1.1. Hadwiger’s Conjecture

Our starting point is Hadwiger’s Conjecture [29], which suggests a deep relationship between graph colourings and graph minors.¹ A *colouring* of a graph G is a function that assigns one colour to each vertex of G . For an integer $k \geq 1$, a *k -colouring* is a colouring using at most k colours. A colouring of a graph is *proper* if each pair of adjacent vertices receives distinct colours. The *chromatic number* $\chi(G)$ of a graph G is the minimum integer k such that G has a proper k -colouring. A graph H is a *minor* of a graph G if H is isomorphic to a graph that can be obtained from a subgraph of G by contracting edges. A graph G is *H -minor-free* if H is

not a minor of G . Let K_h be the complete graph on h vertices. Hadwiger [29] famously conjectured that every K_h -minor-free graph is properly $(h - 1)$ -colourable. This is widely considered to be one of the most important open problems in graph theory. By Wagner’s characterization of K_5 -minor-free graphs [64], the case $h = 5$ of Hadwiger’s Conjecture is equivalent to the 4-Colour Theorem [1, 54]. The conjecture is true for $h \leq 6$ [56], and is open for $h \geq 7$. The best known upper bound on the chromatic number of K_h -minor-free graphs remained of order $O(h\sqrt{\log h})$ [38, 60] since the 1980s, until two recent breakthroughs. First, Norin, Postle, and Song [48] proved an upper bound of $O(h(\log h)^{1/4+\varepsilon})$ for any fixed $\varepsilon > 0$, which was improved by Delcourt and Postle [11] to $O(h \log \log h)$. It is open whether every K_h -minor-free graph is $O(h)$ -colourable. See Seymour’s survey [58] for more on Hadwiger’s Conjecture.

1.2. The Clustered Hadwiger Conjecture

As mentioned above, one of the main ways to approach Hadwiger’s Conjecture has been to try to minimise the number of colours in a proper colouring of a K_h -minor-free graph. A second natural approach is to fix a number of colours close to Hadwiger’s bound (at $h - 1$ for instance), and try to obtain a colouring that is close to being proper. This leads to the notion of improper colourings of K_h -minor-free graphs; see [21, 24, 32, 33, 34, 35, 40, 41, 42, 43, 44, 45, 47, 49, 50, 52, 62, 65] and the references therein. A *monochromatic component* with respect to a colouring of a graph G is a connected component of the subgraph of G induced by all the vertices assigned a single colour. A colouring has *clustering* c if every monochromatic component has at most c vertices. Note that a colouring with clustering 1 is precisely a proper colouring. The *clustered chromatic number* $\chi_*(G)$ of a graph class $\mathcal{G}$ is the minimum integer k for which there exists an integer c such that every graph in $\mathcal{G}$ is k -colourable with clustering c . See [66] for an extensive survey on this topic, and see [3, 27] for connections between clustered colouring and asymptotic dimension in geometric group theory and site percolation in probability theory. One

1. We consider simple, finite, undirected graphs G with vertex-set $V(G)$ and edge-set $E(G)$. See [12] for graph-theoretic definitions not given here.

of the earliest works on clustered colouring was by Kleinberg, Motwani, Raghavan, and Venkatasubramanian [37], who used it as a tool to design algorithms for evolving databases.

Consider the clustered chromatic number of the class of K_h -minor-free graphs. So-called standard examples provide a lower bound of $h - 1$ (regardless of the clustering function); see [Proposition 9](#) below. A natural weakening of Hadwiger's Conjecture, sometimes called the Clustered Hadwiger Conjecture, asserts that every K_h -minor-free graph is $(h - 1)$ -colourable with clustering at most some function $f(h)$. This conjecture was first asked as an open problem by Edwards et al. [24]. This line of research was initiated in 2007 by Kawarabayashi and Mohar [35], who proved the first $O(h)$ upper bound on the number of colours. Their bound was $\lceil \frac{31}{2}h \rceil$, which was successively improved to $\lceil \frac{7h-3}{2} \rceil$ by Wood [65],² to $4h-4$ by Edwards et al. [24], to $3h-3$ by Liu and Oum [41], to $2h-2$ independently by Norin [47], van den Heuvel and Wood [62] and Dvořák and Norin [21], and most recently to h by Liu and Wood [44]. Dvořák and Norin [21] showed that $h - 1$ colours suffice for $h \leq 9$, and Edwards et al. [24] proved that every K_h -minor-free graph has an $(h - 1)$ -colouring in which each monochromatic component has bounded maximum degree (which is significantly weaker than having bounded size).

Our first contribution is to prove the Clustered Hadwiger Conjecture,³ thereby solving the above-mentioned open problem of Edwards et al. [24].

Theorem 1. *Every K_h -minor-free graph is $(h - 1)$ -colourable with clustering at most some function $f(h)$.*

1.3. Excluding a Complete Bipartite Minor

Consider the clustered chromatic number of the class of $K_{s,t}$ -minor-free graphs, where $K_{s,t}$ is the complete bipartite graph with parts of size $s \geq 1$ and $t \geq s$. This question is of particular interest since the answer turns out to not depend on t . Van den Heuvel and Wood [62] proved a lower bound of $s + 1$ (for $t \geq \max\{s, 3\}$; see [Proposition 9](#) below), and observed that results of Edwards et al. [24] and Ossona de Mendez et al. [52] imply an upper bound of $3s$, which was improved to $2s + 2$ by Dvořák and Norin [21], and further improved to $s + 2$ by Liu and Wood [42]. Our second main result resolves the question.

Theorem 2. *Every $K_{s,t}$ -minor-free graph is $(s + 1)$ -colourable with clustering at most some function $f(s, t)$.*

Combined with the above-mentioned lower bound, [Theorem 2](#) shows that the clustered chromatic number of the class of $K_{s,t}$ -minor-free graphs equals $s + 1$. This resolves an open problem proposed by van den Heuvel and Wood [62].

2. The result of Wood [65] depended on a result announced by Norin and Thomas [51, 59], which has not yet been fully written.

3. Dvořák and Norin [21] announced in 2017 that a forthcoming paper, which has not yet been fully written, will also prove the Clustered Hadwiger Conjecture.

1.4. Colin de Verdière Parameter

The Colin de Verdière parameter $\mu(G)$ is an important graph invariant introduced by Colin de Verdière [8, 9]; see [57, 63] for surveys. It is known that $\mu(G) \leq 1$ if and only if G is a disjoint union of paths, $\mu(G) \leq 2$ if and only if G is outerplanar, $\mu(G) \leq 3$ if and only if G is planar, and $\mu(G) \leq 4$ if and only if G is linklessly embeddable. A famous conjecture of Colin de Verdière [8] states that $\chi(G) \leq \mu(G) + 1$, which implies the 4-colour theorem and is implied by Hadwiger's Conjecture. The following clustered analogue was conjectured by Wood [66].

Theorem 3. *The clustered chromatic number of the class of graphs with Colin de Verdière parameter μ equals $\mu + 1$.*

The lower bound in [Theorem 3](#) is proved in [66]. The upper bound follows immediately from either [Theorem 1](#) or [Theorem 2](#) since graphs with Colin de Verdière parameter μ are $K_{\mu+2}$ -minor-free [8, 9] and are $K_{\mu, \max\{\mu, 3\}}$ -minor-free [63].

1.5. A Common Generalization

We in fact prove a common generalization of [Theorems 1](#) and [2](#) using the following definition of Campbell, Clinch, Distel, Gollin, Hendrey, Hickingbotham, Huynh, Illingworth, Tamitegama, Tan, and Wood [7]. Let $\mathcal{J}_{s,t}$ be the set of all graphs $K_s \oplus T$ where T is a t -vertex tree. Here the *complete join* $G_1 \oplus G_2$ is the graph obtained from the disjoint union of graphs G_1 and G_2 by adding all edges between G_1 and G_2 . A graph is $\mathcal{J}_{s,t}$ -minor-free if it contains no graph in $\mathcal{J}_{s,t}$ as a minor.⁴

Theorem 4. *Every $\mathcal{J}_{s,t}$ -minor-free graph is $(s + 1)$ -colourable with clustering at most some function $f(s, t)$.*

Since $K_{s,t}$ is a subgraph of each graph in $\mathcal{J}_{s,t}$, [Theorem 4](#) implies [Theorem 2](#). Since $\mathcal{J}_{h-2,2} = \{K_h\}$, [Theorem 4](#) with $s = h - 2$ implies [Theorem 1](#).

1.6. Excluding a Complete Bipartite Subgraph

As mentioned above, Liu and Wood [42] proved that $K_{s,t}$ -minor-free graphs are $(s + 2)$ -colourable with bounded clustering. They actually proved a stronger result, where the

4. Let $P_{s,t}$ be the graph obtained from $K_{s,t}$ by adding a path on the t -vertex side. Every tree on t^2 vertices contains a $K_{1,t}$ minor or a path on t vertices. It follows that every graph in $\mathcal{J}_{s,t^2}$ contains $K_{s+1,t}$ or $P_{s,t}$ as a minor. Conversely, $P_{s,t+s}$ and $K_{s+1,t+s}$ contain a graph in $\mathcal{J}_{s,t}$ as a minor (obtained by contracting a suitable s -edge matching). This says that (ignoring dependence on t) being $\mathcal{J}_{s,t}$ -minor-free is equivalent to being $K_{s+1,t}$ -minor-free and $P_{s,t}$ -minor-free. Therefore [Theorem 4](#) could be stated for graphs that are $K_{s+1,t}$ -minor-free and $P_{s,t}$ -minor-free. We choose to work with $\mathcal{J}_{s,t}$ for convenience.

number of colours is determined by an excluded complete bipartite subgraph, as expressed in the following results.^{5,6}

Theorem 5 ([42, Theorem 5]). *For any integers $s, t, w \in \mathbb{N}$, every $K_{s,t}$ -subgraph-free graph of treewidth at most w is $(s+1)$ -colourable with clustering at most some function $f(s, t, w)$.*

Theorem 6 ([42, Theorem 2]). *For any integers $t \geq s \geq 1$ and for any graph X , every $K_{s,t}$ -subgraph-free X -minor-free graph is $(s+2)$ -colourable with clustering at most some function $f(s, t, X)$.*

In both these theorems, the number of colours is best possible [42], and we use an extension of [Theorem 5](#) as one of the tools in the current paper. It is worth contrasting our [Theorem 2](#) with [Theorem 6](#). [Theorem 2](#) uses only $s+1$ colours, but applies only to a graph G that contains no $K_{s,t}$ minor. [Theorem 6](#) uses $s+2$ colours, but applies in the more general setting in which G contains no $K_{s,t}$ subgraph and excludes any other (fixed) graph X as a minor.

[Theorem 6](#) implies that K_h -minor-free graphs are $(h+1)$ -colourable with bounded clustering (since K_h is a minor of $K_{h-1, h-1}$). Liu and Wood [44] pushed this proof further to reduce the number of colours to h , as mentioned above. These results are presented over a series of three articles [42, 43, 44]. The main tool introduced in the first article of the series [43] shows (via a technical 70-page proof) that $K_{s,t}$ -subgraph-free graphs of bounded layered treewidth are $(s+2)$ -colourable with bounded clustering. It is open whether $s+1$ colours (which would be best possible) suffice in this setting [42, 43]. Bounded layered treewidth is not a minor-closed property. However, for minor-closed classes, having bounded layered treewidth is equivalent to excluding an apex graph as a minor [20]. Here a graph is *apex* if it can be made planar by deleting at most one vertex. Our next result shows that $s+1$ colours suffice for apex-minor-free $K_{s,t}$ -subgraph-free graphs.

Theorem 7. *For any integers $t \geq s \geq 3$ and for any apex graph X , every $K_{s,t}$ -subgraph-free X -minor-free graph is $(s+1)$ -colourable with clustering at most some function $f(s, t, X)$.*

Several notes on [Theorem 7](#) are in order:

- [Theorem 5](#) is equivalent to saying that $K_{s,t}$ -subgraph-free graphs excluding a fixed planar graph as a minor are $(s+1)$ -colourable with bounded clustering. [Theorem 7](#) strengthens this result to the setting of apex-minor-free graphs (for $s \geq 3$).

5. For a graph X , a graph G is *X -subgraph-free* if no subgraph of G is isomorphic to X .

6. A *tree-decomposition* of a graph G is a collection $(B_x : x \in V(T))$ of subsets of $V(G)$ (called *bags*) indexed by the vertices of a tree T , such that (a) for every edge $uv \in E(G)$, some bag B_x contains both u and v , and (b) for every vertex $v \in V(G)$, the set $\{x \in V(T) : v \in B_x\}$ induces a non-empty (connected) subtree of T . The *treewidth* of a graph G , denoted by $\text{tw}(G)$, is the minimum width of a tree-decomposition of G . Treewidth is the standard measure of how similar a graph is to a tree. Indeed, a connected graph has treewidth at most 1 if and only if it is a tree. See [2, 30, 53] for surveys on treewidth.

- The bound on the number of colours in [Theorem 7](#) is tight, simply because $s+1$ colours is tight for bounded treewidth graphs [42, 43].
- Some non-trivial lower bound on s is needed in [Theorem 7](#), since the hexagonal grid graph is K_5 -minor-free and $K_{1,7}$ -subgraph-free, but every 2-colouring has unbounded clustering by the Hex Lemma [31]. [Theorem 7](#) with $s=2$ is open, although recently solved in the case of graphs of bounded Euler genus by Dvořák [23].
- Our proof of [Theorem 7](#) in the case $s \geq 4$ is reasonably short and simple (see [17]). The case $s=3$ is more difficult, and requires tools for dealing with $K_{2,t}$ -subgraphs in surfaces. These tools are also required by the proofs of [Theorems 1, 2](#) and [4](#).
- Our proofs use and extend results of Liu and Wood [42] for colouring $K_{s,t}$ -subgraph-free graphs of bounded treewidth (including [Theorem 5](#)), but do not use layered treewidth or the 70-page proof of Liu and Wood [43] mentioned above.

1.7. Clustered 4-Colouring Theorems

It is well-known that the clustered chromatic number of the class of planar graphs equals 4, where the upper bound follows from the 4-Colour Theorem or a weaker result of Cowen, Cowen, and Woodall [10], and the lower bound follows from the $s=3$ case of [Proposition 9](#) below. In fact, much more general results are known for 4-colouring with bounded clustering.

The *Euler genus* of a surface with h handles and c cross-caps is $2h+c$. The *Euler genus* of a graph G is the minimum integer $g \geq 0$ such that there is an embedding of G in a surface of Euler genus g ; see [46] for more about graph embeddings in surfaces. Dvořák and Norin [21] proved the following elegant generalization of the 4-Colour Theorem.

Theorem 8 ([21]). *Every graph of Euler genus g has a 4-colouring with clustering at most some function $f(g)$.*

It follows from Euler's formula that graphs of Euler genus g are $K_{3,2g+3}$ -minor-free. Thus the $s=3$ case of [Theorem 2](#) generalises [Theorem 8](#) to the setting of $K_{3,t}$ -minor-free graphs. This setting is substantially more general since disjoint unions of K_5 are $K_{3,3}$ -minor-free, but have unbounded Euler genus. This highlights the utility of considering excluded complete bipartite minors.

[Theorem 7](#) with $s=3$ provides a further generalization where the $K_{3,t}$ -minor-free assumption is relaxed to apex-minor-free and $K_{3,t}$ -subgraph-free. Again, this is a substantial generalization since, for example, the 1-subdivision of $K_{3,n}$ is K_5 -minor-free and $K_{3,3}$ -subgraph-free, but contains a $K_{3,n}$ minor.

Dvořák and Norin's proof of [Theorem 8](#) uses the so-called 'island' method. Our proof uses and builds on this approach.

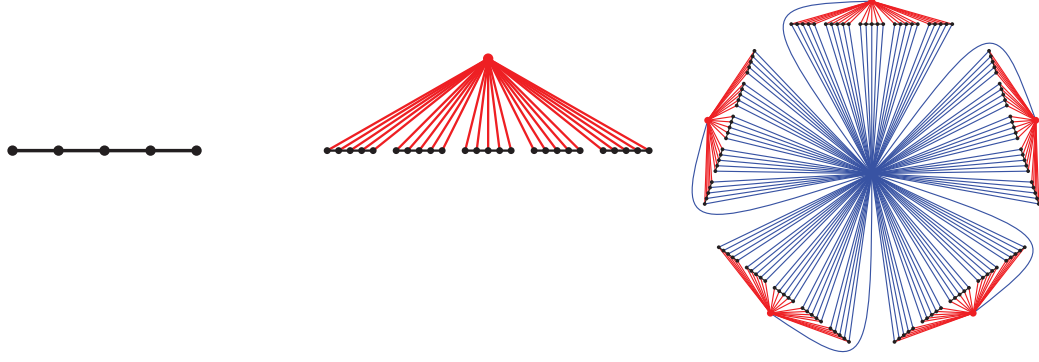


Figure 1. Members of the families $\mathcal{G}_1$, $\mathcal{G}_2$, and $\mathcal{G}_3$.

1.8. Lower Bounds

The number of colours in Theorems 1, 2 and 7 is best possible because of the following well-known ‘standard’ example [24, 39, 42, 62, 66], which we include for completeness.

Proposition 9. *For each integer $s \geq 1$, there is a graph class $\mathcal{G}_s$, such that every graph in $\mathcal{G}_s$ has treewidth s and is $K_{s,s+2}$ -subgraph-free, and for any integer $c \geq 1$ there exists $G \in \mathcal{G}_s$ such that every s -colouring of G has a monochromatic component on more than c vertices.*

Proof. We proceed by induction on s , where $\mathcal{G}_1$ is the class of all paths on at least two vertices, which trivially satisfies the desired properties. Now assume the claim for some $s \geq 1$. As illustrated in Figure 1, let $\mathcal{G}_{s+1}$ be the class of all graphs obtained by taking arbitrarily many disjoint copies of graphs in $\mathcal{G}_s$ and adding one dominant vertex.

Every graph in $\mathcal{G}_{s+1}$ has treewidth $s+1$ and is $K_{s+1,s+3}$ -subgraph-free. For any c , there is a graph H in $\mathcal{G}_s$ such that any s -colouring of H has a monochromatic component on more than c vertices. Let G be obtained from c disjoint copies of H by adding a dominant vertex v . So $G \in \mathcal{G}_{s+1}$. In any colouring of G with clustering c , some copy of H avoids the colour assigned to v , as otherwise the monochromatic component containing v would have at least $c+1$ vertices. By assumption, H has at least $s+1$ colours, implying G has at least $s+2$ colours. This proves the claim. $\square$

Proposition 9 shows that $h-1$ colours in Theorem 1 is best possible, since every graph with treewidth s is K_{s+2} -minor-free. Similarly, Proposition 9 shows that $s+1$ colours in Theorem 2 is best possible, since the graph in Proposition 9 is $K_{s,\max\{s,3\}}$ -minor-free [62]. To see that the number of colours in Theorem 7 is best possible, note that every graph in $\mathcal{G}_s$ contains no $(s+1) \times (s+1)$ grid as a minor (since the $(s+1) \times (s+1)$ grid has treewidth $s+1$). Thus $\mathcal{G}_s$ excludes a planar (and thus apex) graph as a minor. If every graph in $\mathcal{G}_s$ is k -colourable with clustering at most some function $f(s)$, then $k \geq s+1$ by Proposition 9.

2. Proof Outline

This section gives a high-level sketch of the proof of Theorems 4 and 7. Full proofs are available in reference [17]. Here we highlight the main challenges that arise when adapting existing techniques to prove our results, and give a rough idea of how we overcome these challenges. The key to our proofs is the novel use of ‘graph product structure theory’ in partnership with the Graph Minor Structure Theorem of Robertson and Seymour [55].

Graph product structure theory is a recently developed field that describes graphs in a complicated graph class as subgraphs of the strong product (sometimes along with some other operations) of simpler graphs which typically have bounded treewidth. The *strong product* of graphs A and B , denoted by $A \boxtimes B$, is the graph with vertex-set $V(A) \times V(B)$, where distinct vertices $(v, x), (w, y) \in V(A) \times V(B)$ are adjacent if $v = w$ and $xy \in E(B)$, or $x = y$ and $vw \in E(A)$, or $vw \in E(A)$ and $xy \in E(B)$.

The starting point for these recent developments is the Planar Graph Product Structure Theorem of Dujmović, Joret, Micek, Morin, Ueckerdt, and Wood [19], which says that every planar graph G is a subgraph of the strong product of a graph H of treewidth 8 and a path P , written as $G \subseteq H \boxtimes P$. (The bound of 8 was reduced to 6 by Ueckerdt, Wood, and Yi [61].) This result often allows a question about planar graphs (G) to be reduced to an analogous question about graphs of bounded treewidth (H), which is usually easier to solve. Extending the solution for H to the product $H \boxtimes P$ is often straightforward. This approach has been used to solve several decades-old problems in mathematics and theoretical computer science, including bounding the queue-number of planar graphs [19], bounding the nonrepetitive chromatic number of planar graphs [15], the construction of nearly optimal adjacency labelling schemes for planar graphs [14, 25], amongst other results [4, 5].

This approach is, of course, limited to graph families that admit this type of product structure. Dujmović et al. [19] proved that apex-minor-free graphs are the most general minor-closed classes contained in the product of a bounded treewidth graph and a path, so this reduction cannot be

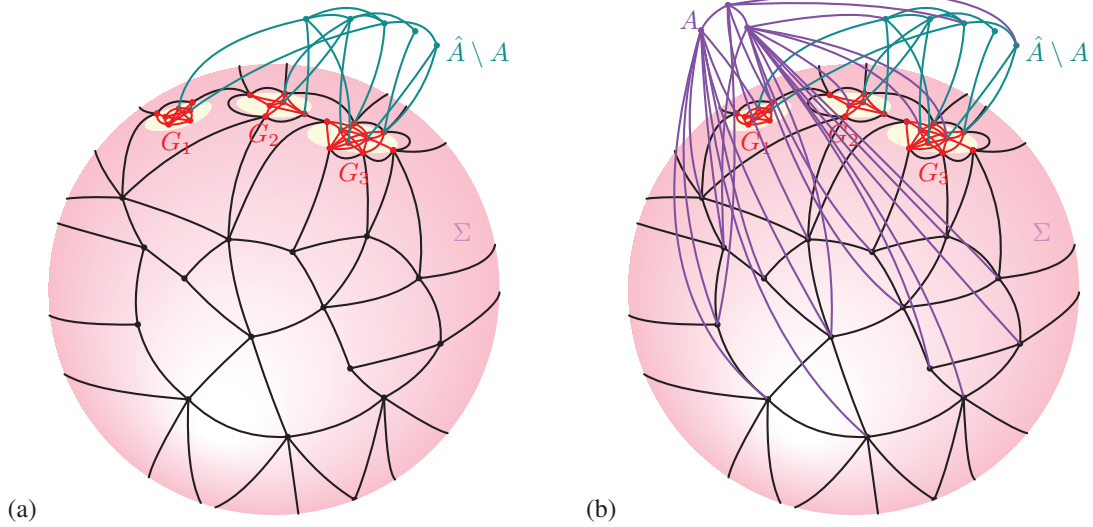


Figure 2. (a) An almost-embedded graph with no major apex vertices. (b) An almost-embedded graph with three major apex vertices. The discs used to define vortices are yellow. Edges and vertices that participate in vortices are red. Non-major apex vertices and their incident edges are dark cyan. Major apex vertices and their incident edges are purple. Non-top edges and vertices are black.

applied to the graph classes considered by Theorems 1, 2 and 4 (except for a few small values of h or s).

Since not all minor-closed classes admit the type of product structure described above, we start by applying the Graph Minor Structure Theorem of Robertson and Seymour [55], which describes graphs excluding a fixed minor in terms of graphs called ‘torsos’ that are formed from a surface-embedded subgraph, by adding vortices in the surface, and apex vertices with unrestricted neighbourhoods. The torsos are then pasted together in a tree-like way (described by a tree-decomposition) using clique-sums.

A natural strategy for colouring a $\mathcal{J}_{s,t}$ -minor-free graph is to colour each torso one-by-one beginning with the root torso. With this strategy, when colouring a subsequent non-root torso, the vertices it shares with its parent torso are already precoloured with colours assigned while colouring the parent torso. To control the growth of monochromatic components, we demand that the colouring of the child torso ‘properly extends’ the precolouring, in the sense that there is no monochromatic edge uv with u precoloured and v not precoloured. The precoloured vertices are a clique-sum whose size is bounded by a function of s and t , but this bound may be much larger than s , which is a major obstacle since Theorem 4 promises to use at most $s + 1$ colours. Indeed, imagine that there are $s + 1$ precoloured vertices assigned distinct colours and there is one uncoloured vertex v adjacent to all $s + 1$ precoloured vertices. Then no colour is available for v since we require that the new colouring properly extends the given precolouring (since there are only $s + 1$ colours available). This motivates the following definition.

Call a clique-sum *small* if it has at most s vertices, and *large* otherwise (although still of size bounded by a function of s and t). Using the approach described above,

the precoloured vertices of torsos are either part of a small or a large clique-sum. Since the natural strategy is not applicable to large clique-sums we conduct those clique-sums before any colouring. In particular, we call a maximal collection of torsos pasted together by only large clique-sums a *curtain*. (This choice of name will become clear later.) Now one can think of the Robertson–Seymour decomposition as a collection of curtains glued together using small clique-sums, which we call a *tree of curtains*. To colour the entire graph, it suffices to colour each curtain one-by-one beginning with the curtain that contains the root torso. The critical advantage of proceeding this way is that each non-root curtain comes with at most s precoloured vertices.

To colour each individual curtain, we use a variant of the Graph Minor Structure Theorem by Dvořák and Thomas [22], which classifies apex vertices as either *major* or *non-major*. As illustrated in Figure 2, edges incident to *major* apex vertices are unrestricted, but *non-major* apex vertices are only adjacent to vortices and other (major and non-major) apex vertices. A graph is *k-apex* if it has k vertices whose removal leaves a planar graph. Dvořák and Thomas [22] showed that for k -apex-minor-free graphs, the number of major apex vertices is at most $k - 1$. In our case, $\mathcal{J}_{s,t}$ includes $K_s \oplus P_t$, which is $(s - 2)$ -apex since $K_2 \oplus P_t$ is planar. So the number of major apex vertices is at most $s - 3$. From now on we group the non-major apex vertices with the vortices (loosely speaking).

An important consequence of the Dvořák and Thomas [22] result is that (after some manipulation) any large clique-sum only involves (major and non-major) apex vertices and vertices from vortices of the two torsos being summed. Therefore, none of the clique-sums used to make a curtain touch the surface-embedded parts of the torsos, except those vertices on the boundary of a vortex.

Our task now is to colour each curtain, given a set of at most s precoloured vertices. Recall that each torso in the curtain is described by a set of at most $s - 3$ major apex vertices, a surface-embedded subgraph, and a collection of vortices and non-major apex vertices. We now apply the so-called ‘island’ method of Esperet and Ochem [26] and Dvořák and Norin [21]. A *d-island* is a set of vertices, each of which has at most d neighbours outside I . If the surface-embedded subgraph contains a 3-island I disjoint from the vortices and of bounded size, then it is an s -island in the overall graph (including the at most $s - 3$ major apex vertices as possible neighbours). Delete I , apply induction, and greedily colour each vertex in I by a colour not used on the at most s neighbours outside I . Each new monochromatic component is contained within I , and thus has bounded size. Now we may assume the surface-embedded subgraph of each torso (within a curtain) has no 3-island of bounded size disjoint from the vortices.

The next step employs graph product structure theory. A result of Dujmović et al. [19] says that each torso (without the major apex vertices) is a subgraph of the strong product of a bounded treewidth graph and a path (generalising the result for planar graphs mentioned above). At this point it is helpful to view this product structure in terms of partitions and layerings, which we now introduce.

A *partition* of a graph G is a collection $\mathcal{P}$ of non-empty sets of vertices in G such that each vertex of G is in exactly one element of $\mathcal{P}$. Each element of $\mathcal{P}$ is called a *part*. The *quotient* of $\mathcal{P}$ is the graph, denoted by $G/\mathcal{P}$, with vertex set $\mathcal{P}$ where distinct parts $A, B \in \mathcal{P}$ are adjacent in $G/\mathcal{P}$ if and only if some vertex in A is adjacent in G to some vertex in B . Partitions and clustered colouring are intimately related, since a graph G is k -colourable with

clustering c if and only if G has a partition $\mathcal{P}$ such that every part of $\mathcal{P}$ has at most c vertices and the quotient $G/\mathcal{P}$ is properly k -colourable. A *layering* of a graph G is an ordered partition $\mathcal{L} = (L_1, L_2, \dots)$ of $V(G)$ such that for every edge $vw \in E(G)$, if $v \in L_i$ and $w \in L_j$, then $|i - j| \leq 1$. It follows from the definitions that for any graph H and path P , a graph G is isomorphic to a subgraph of $H \boxtimes P \boxtimes K_c$ if and only if G has a partition $\mathcal{P}$ and a layering $\mathcal{L}$ such that $|A \cap L| \leq c$ for each part $A \in \mathcal{P}$ and each layer $L \in \mathcal{L}$.

By the above-mentioned result of Dujmović et al. [19], for each torso with the major apex vertices removed, we obtain a partition of the vertex-set with connected parts, and a layering such that the intersection of each part and each layer has bounded size (thus, loosely speaking, each part in the partition is ‘long and skinny’). In addition, the first layer contains exactly the vortices and the non-major apex vertices. This means that each large clique-sum used to paste together a pair of torsos in the curtain G is restricted to the major apex vertices and the vertices in the first layer of each torso. Dujmović et al. [19] proved that the minor obtained by contracting each part of the partition results in a quotient graph that has bounded treewidth.

However, we do not perform the contractions that would create this bounded treewidth quotient graph, since doing so could interfere with a clique-sum and could introduce complete bipartite subgraphs that we wish to avoid for reasons that will become clear shortly. Instead, we reduce to a bounded treewidth graph by ‘raising the curtain’ as follows; see Figure 3. Within each torso of a curtain G , we contract the parts of the corresponding partition while avoiding the first five layers. This produces a minor $G_\uparrow$ of G on at most six layers and having bounded treewidth. Since

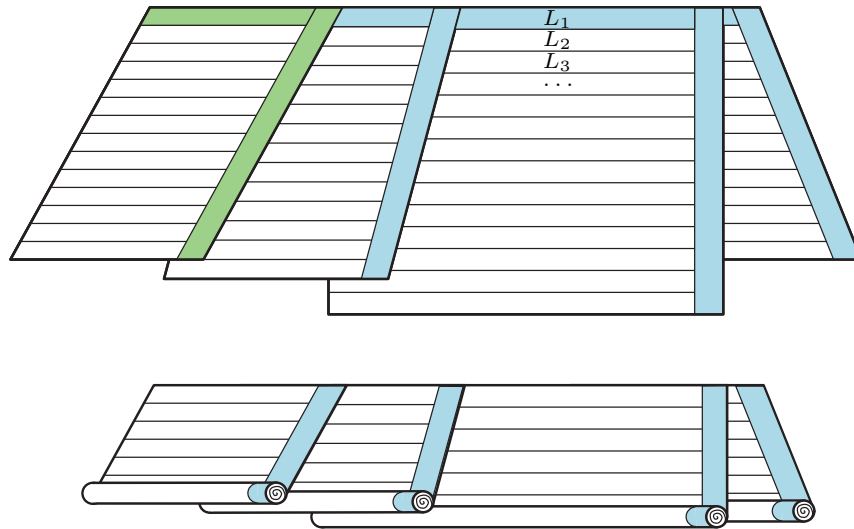


Figure 3. A curtain (above) and a raised curtain (below). We think of the layers as drawn from top to bottom (with the top of the root torso of the curtain being depicted in green). All torsos of the decomposition are glued on a subset of their first and second layers (and on a subset of their major apex vertices, which are depicted in blue—or green, for the major apex vertices A_r of the root torso—on the figure and are not part of the layerings).

these contractions avoid the first five layers of each torso, they avoid the vertices used in the clique-sums between torsos. Therefore, these clique-sums remain in $G_{\uparrow}$, and $G_{\uparrow}$ is also a curtain with a surface-embedded subgraph within each torso. Unlike G , $G_{\uparrow}$ has bounded treewidth.

However, it is not enough that $G_{\uparrow}$ has bounded treewidth. Indeed, the extremal examples have bounded treewidth (see Proposition 9). It is also critical that $G_{\uparrow}$ is a minor of G . However, because of the possibility of edges added to each torso, it is not immediate that $G_{\uparrow}$ is obtained by contracting connected subgraphs in G . To overcome this issue we show that our tree of curtains is ‘lower-minor-closed’. We expect this property may be of independent interest.

We prove two lemmas (that are technical extensions of results by Liu and Wood [42, 44]) for colouring bounded treewidth graphs containing no $K_{s,t}$ -subgraph. Our goal is to apply these results to $G_{\uparrow}$. However, without further work $K_{s,t}$ -subgraphs may be present in $G_{\uparrow}$.

Any such problematic subgraph contains a large $K_{2,p}$ -subgraph in the deeper layers of some surface-embedded subgraph of $G_{\uparrow}$. We eliminate these problematic $K_{2,p}$ -surface-embedded subgraphs in two steps: (i) We do further contractions of connected surface-embedded subgraphs of $G_{\uparrow}$ that are separated from the rest of $G_{\uparrow}$ by short surface-separating cycles and major apex vertices. (ii) We remove certain redundant surface degree-2 vertices. The result of this process is a minor $G_{\uparrow\bullet}$ of $G_{\uparrow}$ that shares a crucial property with $G_{\uparrow}$: Each vertex of $G_{\uparrow\bullet}$ corresponds to a long and skinny connected subgraph of G . (Establishing this property relies on the fact that 3-islands have been previously removed from each surface-embedded subgraph.)

We augment $G_{\uparrow\bullet}$ with a set of ‘special’ vertices, each of which dominates a connected subgraph in the embedded part of some torso of $G_{\uparrow\bullet}$, as illustrated in Figure 5. The addition of these special vertices only increases the treewidth by a small amount. On this augmented graph we apply

an enhanced version of the divide-and-conquer strategy for $K_{s,t}$ -subgraph-free bounded treewidth graphs of Liu and Wood [42, 44] to find an $(s + 1)$ -colouring with bounded clustering where, for each special vertex α , the subgraph dominated by α completely avoids the colour used by α .

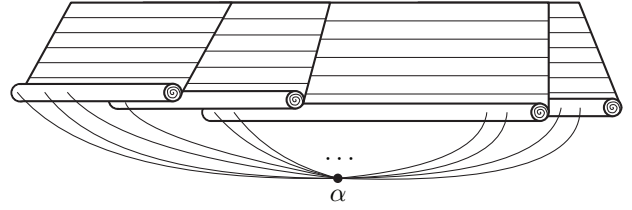


Figure 5. Adding the vertex α adjacent to the bottom layer of $G_{\uparrow\bullet}$.

We now lower the curtain that was previously raised, and assign each vertex w of G the colour given to the vertex of $G_{\uparrow\bullet}$ that w was contracted into. The resulting colouring of G does not have bounded clustering, but its monochromatic components are all long and skinny (with respect to the layering and partition). As illustrated in Figure 4, this allows us to use the colours assigned to the special vertices α as blocking colours to break any long and skinny monochromatic components into (short and skinny) pieces of bounded size.

In this way, we obtain an $(s + 1)$ -colouring of the curtain G with bounded clustering (properly extending a precolouring of the at most s vertices in the parent torso). Then we use the top-down colouring strategy mentioned earlier to extend this colouring to an $(s + 1)$ -colouring of the tree of curtains, which is the original $\mathcal{J}_{s,t}$ -minor-free graph.

This completes the high-level description of the proof of Theorem 4.

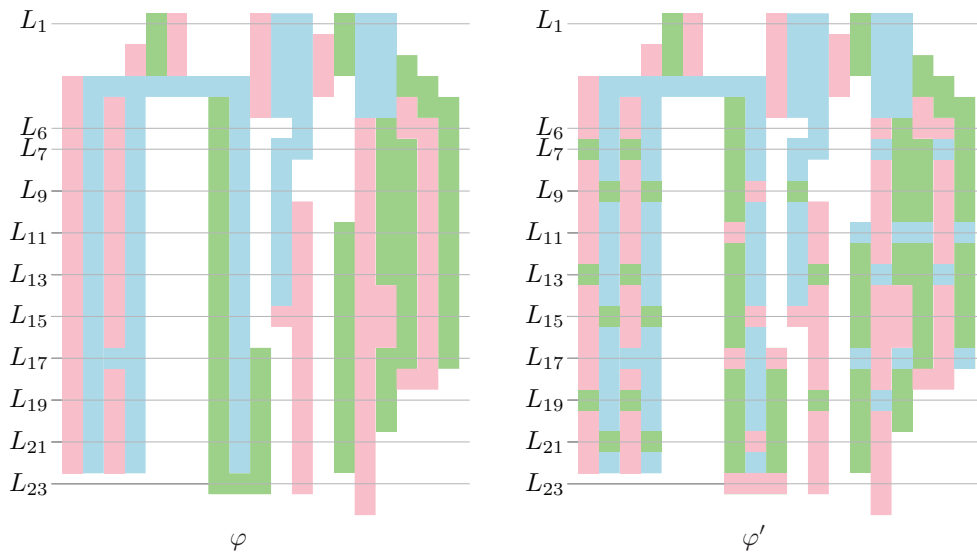


Figure 4. Breaking long and skinny monochromatic components in the 3-colour case.

3. Conclusion

The proofs of all our results are constructive and yield polynomial $O(n^c)$ time colouring algorithm, where c is independent of the excluded minor or excluded subgraph. Here we use known polynomial-time algorithms [21, 28, 36] for finding the decomposition in the Graph Minor Structure Theorem as a starting point, and we use existing algorithms for finding the product structure description of an almost-embedded graph without major apex vertices [6, 19].

Theorems 1 and 2 give the first optimal bounds on the clustered chromatic number of K_h -minor-free and $K_{s,t}$ -minor-free graphs. At least as important as these results are the definitions and tools that we develop, including curtain decompositions, raised curtains, and skinniness-preserving contractions to control $K_{2,t}$ subgraphs. Curtain decompositions have already found applications to other decomposition and colouring problems [18] and we expect that they will soon find more. Since the extremal examples for many problems include large $K_{2,t}$ subgraphs, we expect the tools for controlling $K_{2,t}$ -subgraphs will also find further applications in different contexts. Indeed, the Planar Graph Product Structure Theorem mentioned above was initially developed to bound the queue-number of planar graphs [19], but has since been used to resolve a number of longstanding open problems on planar and other graph classes [4, 5, 13, 14, 15, 16].

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